

Long Hair Identification System

1) Project Overview

This project extends the Car Colour Detection and Traffic Analysis training project by implementing an Age-Gated Long Hair Identification System.

The system combines Deep Learning and Computer Vision techniques to:

1. Predict a person's age from a facial image.
2. Predict a person's gender using a CNN.
3. Detect hair length (Long or Short) using image processing.
4. Apply an age-gated rule: for predicted ages 20-30, gender is decided by hair length instead of the CNN's own prediction.
5. Provide a graphical user interface (GUI) for image upload and result visualization.

2) Problem Statement

Standard face-analysis systems predict gender directly from a photo using a CNN. This task adds a deliberate age-dependent override rule on top of that baseline behaviour.

The objective of this project is to develop a machine learning system capable of:

1. Predicting age and gender from a person's photo using two CNNs.
2. Detecting hair length using simple image processing, since no public dataset provides hair-length labels.
3. Overriding the gender prediction based on hair length for people whose predicted age falls between 20 and 30.
4. Displaying results through a user-friendly GUI.

3) Dataset

UTKFace Dataset

Dataset Source:

Kaggle - jangedoo/utkface-new

Labels are embedded directly in each filename, so no separate annotation step was required:

`[age]_[gender]_[race]_[timestamp].jpg`

Here, gender 0 = Male and 1 = Female.

Dataset Structure:

- 4,000 images used (a subset of the ~23,000 available) to keep training fast
- 3,200 training images / 800 testing images (80/20 split)
- 1,328 of the 4,000 images (about 33%) fall inside the 20-30 age gate

4) Methodology

1. Data Preprocessing

- Filenames parsed to extract age and gender labels.
- Images resized to 64 x 64 pixels.
- Pixel values normalized to [0,1] using a Rescaling layer inside the model.
- Dataset split into training and testing sets (80/20).

2. Gender Classification Model

A Convolutional Neural Network (CNN) was trained to classify gender, built the same way as the vehicle colour CNN in the training project: a stack of Conv2D + MaxPooling2D blocks, then Dense layers.

Model Architecture:

- Convolution Layer (32 filters)
- Max Pooling Layer
- Convolution Layer (64 filters)
- Max Pooling Layer
- Convolution Layer (128 filters)
- Max Pooling Layer
- Dense Layer (128 units)
- Dropout Layer (0.3)
- Dense Output Layer (1 unit, Sigmoid)

3. Age Estimation Model

Instead of designing a new architecture, this reuses the exact same CNN block structure as the gender model above. Only the final layer changes, from a Sigmoid (yes/no) output to a Linear (numeric) output, since age is a continuous value rather than a category.

4. Hair Length Detection

No face dataset comes with hair-length labels, so instead of training a third model, hair length is estimated with simple, rule-based image processing - the same kind of logic used for the blue/other vehicle colour check in the training project, applied to a different signal:

1. Detect the face using OpenCV's Haar cascade face detector.
2. Examine the region directly below the detected face (roughly chin to shoulder height).
3. Count the fraction of pixels in that region that are dark (pixel value below a fixed threshold of 60).
4. If the dark-pixel ratio is above 0.25, classify hair as "Long"; otherwise, "Short".

5. Age-Gated Long Hair Rule

For each processed image:

1. Predict age and gender using the two CNNs.
2. Detect hair length using the rule above.
3. If the predicted age is between 20 and 30:
 - o Long hair → Female
 - o Short hair → Male

4. Else:

- o Use the CNN's own gender prediction, unaffected by hair length.

5) GUI Development

Gradio was used to develop a graphical user interface.

Features:

- Upload a person image
- Preview the uploaded image
- Predict age using the age CNN
- Predict gender using the gender CNN
- Detect hair length
- Apply the age-gated rule
- Display the final predicted gender alongside all intermediate results

6) Results

Gender Classification

Test Accuracy: 86 %

Male F1-score: 0.88

Female F1-score: 0.85

Full classification report:

- Male - Precision: 0.83, Recall: 0.93, F1-score: 0.88, Support: 415
- Female - Precision: 0.91, Recall: 0.79, F1-score: 0.85, Support: 385
- Overall Accuracy: 0.86 (800 test images)

Age Estimation

Test MAE: 9.79 years

Age-Gated Rule Verification

Successfully verified on:

- Hand-picked logic test cases covering every branch of the required behaviour (in-range/out-of-range, long/short hair, model agreeing or disagreeing with the hair-length verdict).
- Real dataset images, confirming the CNN's own gender prediction is genuinely overridden for ages 20-30, not just coincidentally matching.

Example: a real UTKFace image (true age 28, true gender Male, file 28_0_3_...jpg) produced Predicted Age 27.3, Model's Gender Prediction Male, Hair Length Long, Age Gate Applied Yes, and Final Gender Female - confirming the override fires correctly.

7) Visualizations

Included in notebook/repository:

- Age Distribution Histogram

- Gender Distribution Bar Chart
- Gender Model Training Accuracy Graph
- Gender Model Training Loss Graph
- Gender Model Confusion Matrix
- Age Model MAE Graph
- Age Model Training Loss Graph
- Predicted vs Actual Age Scatter Plot
- Sample Predictions Grid (age, hair length, model gender vs. final gender)

8) Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- Matplotlib
- Scikit-Learn
- Gradio

9) Future Improvements

- Train on more images (this notebook uses a subset of ~4,000 for speed).
- Use data augmentation.
- Improve age estimation with transfer learning.
- Train a dedicated hair-length classifier instead of a fixed dark-pixel threshold.
- Improve face/hair segmentation.
- Add confidence scores to predictions.
- Combine this with the earlier YOLOv8 pipeline to run age/gender/hair detection on every face in a group photo.
- Deploy the application using Hugging Face Spaces, Streamlit, or Flask.

10) Conclusion

This project successfully extends the Car Colour Detection and Traffic Analysis training project with a new Deep Learning feature: an age-gated gender prediction system. The developed system predicts age and gender from a face photo, detects hair length using rule-based image processing, applies the required age-gated override for individuals aged 20-30, and displays results through an interactive GUI. The override behaviour was verified against both hand-picked logic tests and real dataset examples, confirming it works exactly as specified.